

Supratim Nag

Howrah, West Bengal, India

(+91) 8420945255 | snsupratim@gmail.com | <https://github.com/snsupratim> | [snsupratim-portfolio](https://www.linkedin.com/in/snsupratim) | <https://www.linkedin.com/in/snsupratim>

Summary

Skilled in full-stack development with expertise in MERN, Python, JavaScript, and AI/ML. Experienced in building production-ready web apps, RESTful APIs, and GenAI models. Quick learner with a strong grasp of backend logic, version control, and collaborative project execution in hybrid teams.

Education

MCKV Institute of Engineering

BTech in CSE with specializing in AI/ML

• 9.3/10 CGPA

Howrah, West Bengal, India

October 2022 - June 2026*

Technical Skills

Programming Languages : Java, Javascript, Python, php, TypeScript.

Web Libraries/Frameworks: Spring Boot, Flask, React.js, Node.js, Express.js, NextJs, FastAPI, Streamlit, Tensorflow, PyTorch, LangChain Community.

DevOps & Tools : Git, GitHub, Docker (basic), CI/CD (basic), Render, MLOps

Database Systems: MySQL, MongoDB, VectorDB(Pinecone, Chroma)

Internships / Experience

Vaidsys Technologies

AI Intern

Remote

20th May 2025 - 19th June 2025

- Designed and implemented an end-to-end sentiment analysis system using Python and Machine Learning.
- Built a multi-class sentiment classifier (Positive, Negative, Neutral, Irrelevant) using Random Forest and TF-IDF-based NLP features.
- Developed and deployed a Flask REST API for real-time feedback ingestion and stored processed data in MongoDB Atlas.
- Ensured seamless integration of model inference with backend services to support scalable deployment.

Exposys Data Labs

Data Science Intern

Remote

27th April 2025 - 30th May 2025

- Developed a supervised machine learning model to predict early onset of diabetes, achieving ~90% accuracy for real-time health risk assessment.
- Processed and cleaned medical datasets using Pandas and NumPy; applied feature selection and outlier removal techniques to improve model performance.
- Utilized scikit-learn to implement classification models and evaluate metrics such as accuracy, precision, and recall.

Bluestock Fintech

Software Engineering Intern

Remote

8th March 2025 - 28th April 2025

- Co-led a team of 8 interns in developing a full-stack IPO Management System, ensuring timely delivery aligned with functional requirements.
- Designed and implemented secure, RESTful APIs using the MERN stack to manage IPO-related entities with robust CRUD functionality.
- Integrated user authentication and role-based access control to streamline workflows for clients, admins, and investors.

Projects

Crop-Disease Detection

Detection of various crop disease using ResNet-50.

Pytorch, Flask, MERN Stack

Github

- Trained a model using ResNet-50 to be able to detect various crop leaf disease accurately.
- An simple web application to detect crop disease by uploading infected crop leaf images.
- This module is a integral part of a project as "AgriVision" that inend to help farmer in their farming process.

RBAC Modular RAG Medical Assistant

Build an AI Chatbot with LLMs (Groq - Llama3)

Python, FastAPI, MongoDB, Render, Streamlit

Github

- Developed a **role-based AI healthcare assistant** for doctors, nurses, and patients to interact with a medical knowledge base and obtai **domain-specific responses**.
- Implemented **Pinecone** for vector-based retrieval, **Google AI Embeddings** for semantic search, and **Groq API** for high-speed, context-aware response generation.
- Designed a **modular RBAC architecture** with **FastAPI backend** for authentication/authorization, **MongoDB Cloud** for secure storage of user data and chat history, and **Streamlit frontend** for interactive role-based querying.

Shop Assistant RAG ChatBot

Build an RAG Powered Chatbot with Gemini AI

Python, FastAPI, Streamlit, MySQL

Github

- Developed an **AI-powered Retrieval-Augmented Generation (RAG) chatbot** to assist customers in browsing and querying a **shop's product catalog**.
- Integrated **Pinecone Vector DB** for semantic search and **Gemini AI** for natural language responses.
- Implemented **Flask backend** for data processing and **Streamlit frontend** for interactive user experience.
- Designed MySQL-based structured storage for product details alongside vector embeddings for efficient retrieval.

Training & Courses

Industrial Exposer

Hyland - Trainee | 2024

- Participated in an in-depth industrial exposure program at Hyland, gaining valuable insights into the workings of software industry.
- Acquired hands-on understanding of the end-to-end software development lifecycle — from initial requirement gathering and design to development, testing & deployment.

Java Spring Boot Development

MCKV Institute of Engineering| 2025

- Successfully completed a comprehensive academic course covering **Java**, **Spring Boot**, **JPA**, and **Hibernate ORM**.
- Built multiple backend modules using RESTful APIs, integrated databases with Hibernate, and applied JPA for efficient data persistence.

Certifications

Database Management System - Part1&2

Infosys Springboard | 2025

- Completed a certified course focused on Oracle RDBMS and core SQL operations.

Python for Data Science

Udemy | 2024

- Completed a comprehensive course covering data analysis, visualization, and machine learning using Python.

Competitions & Hackathon

Solution for Healthcare Data Interoperability :

Digital Health Symposium at IIT Kharagpur, 2023

Addressing Data Interoperability

- Presented a poster at the IIT Kharagpur event to explain my solution how to enhance data interoperability in Electronic healthcare data.

Project AgriVision

Eco Hackthon at IIT(BHU) Varanasi, 2025

Enhancing Agricultural Productivity and Sustainability through Resource Optimization and Disease Detection

- Presented a compact solution for farmers to enhance their farming productivity through various aspects.
- Online Marketplace for farmers to sell & buy farming products.
- Crop disease detection model ,Yield prediction with NDVI feature to educate farmers about their land and vegetation.